

4. System Analysis

System Analysis is the process of examining a business situation and identifying the key components, data flows, and interactions required to construct an information system. This chapter details the actors, use cases, data flow diagrams (DFD Level-0 and Level-1), database entity models, and system architecture.

4.1 Actors Table

An actor represents any entity (human user or external system) that interacts with the system to perform a function.

Actor Name	Type	Description / Role in System
Customer	Human Actor	Browses products, manages the shopping cart, places orders, and views order history.
Vendor	Human Actor	Registers a shop, uploads and manages products, views and ships customer orders, and monitors shop analytics.
Admin	Human Actor	Platform administrator who approves/rejects vendor accounts, manages product categories, views and manages all products, all orders, all users and customers, and monitors platform-wide statistics and revenue.
System	Automated Actor	Performs backend calculations, enforces JWT tokens, handles database transactions, and manages inventory levels automatically.

4.2 Use Case Diagram

The Use Case diagram visualizes the relationships between actors and the system's core capabilities. The system boundary (dashed rectangle) encloses all platform processes.

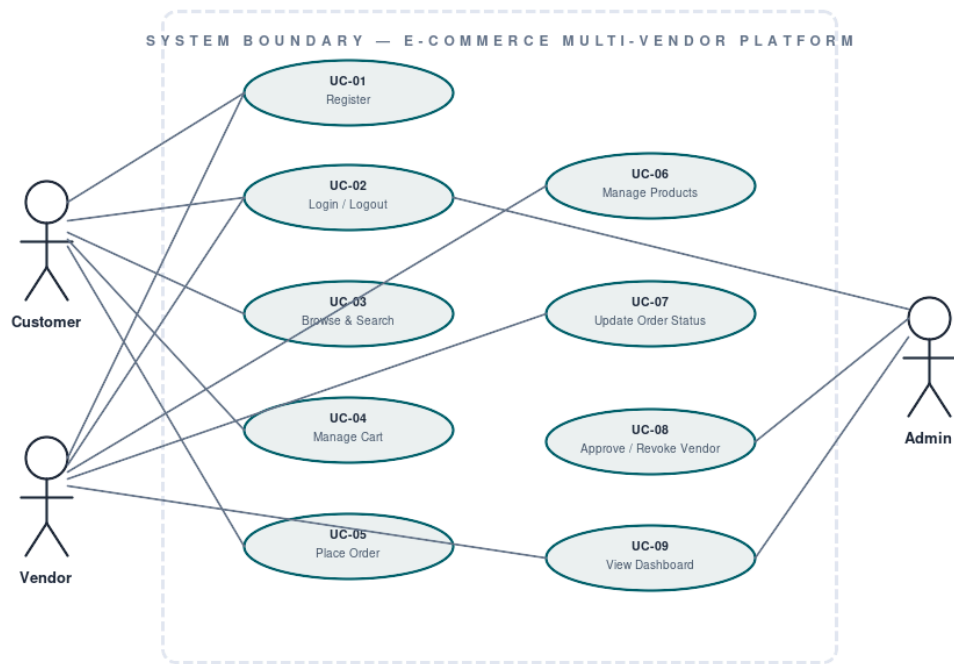


Figure 1 — Use Case Diagram showing Customer, Vendor, and Admin actors

4.3 Use Case Specifications

Below are the structured specifications for the core use cases of the platform.

Use Case ID	Detailed Description
UC-01: User Registration	Primary Actor: Customer / Vendor Description: Allows a guest user to create an account by specifying username, email, password, and their specific role. Precondition: User is not authenticated and has a unique email. Postcondition: A new User record is created in the

	database (along with an empty, unapproved Vendor profile if role was vendor).
UC-02: User Login	Primary Actor: All Roles (Customer, Vendor, Admin) Description: Authenticates the user with email/password and returns a JWT token pair. Precondition: Registered account exists. Postcondition: Browser stores JWT access and refresh tokens; user redirected to their role-specific dashboard.
UC-03: Browse & Search	Primary Actor: Customer / Guest Description: Fetches all active products, allowing filter by category and search by keyword. Precondition: None (Public access). Postcondition: Dynamic list displays matched product cards with stock status.
UC-04: Manage Cart	Primary Actor: Customer Description: Adds items to cart, updates quantities, and calculates rolling totals. Precondition: Customer is logged in. Postcondition: Cart state is updated in memory and local storage.
UC-05: Place Order	Primary Actor: Customer / System Description: Submits cart contents, checks inventory, saves order rows, and deducts item stock in a single database transaction. Precondition: Active cart containing items, valid delivery address, user logged in. Postcondition: Order and OrderItem records created, cart cleared, product stock reduced.
UC-06: Manage Products	Primary Actor: Vendor (Approved) Description: Adds, edits, and deletes products in their storefront catalog. Precondition: Vendor is approved and authenticated. Postcondition: Catalog changes are reflected in the database and public view.

UC-07: Update Order Status	Primary Actor: Vendor Description: Adjusts order status (Pending → Shipped → Delivered) for items belonging to their shop. Precondition: Order containing vendor's products has been placed. Postcondition: Order status updated in the database.
UC-08: Approve / Revoke Vendor	Primary Actor: Admin Description: Toggles the approval status for a vendor shop, enabling or blocking product uploads. Precondition: Admin is logged in, targeted vendor account exists. Postcondition: Vendor's <code>is_approved</code> boolean toggled in database.
UC-09: View Dashboard	Primary Actor: All Roles Description: Displays role-specific statistics and summaries (orders, revenue, users, products) fetched from <code>/api/dashboard/stats/</code>. Precondition: User is authenticated. Postcondition: Dashboard statistics are fetched from API and rendered on screen.
UC-10: Manage Profile	Primary Actor: All Roles Description: Any authenticated user can view and update their profile (username, email, phone, address). Vendors additionally update their shop name and description via a PATCH to <code>/api/auth/me/</code>. Precondition: User is authenticated. Postcondition: Profile details updated in the database and reflected in the AuthContext state.

4.4 Data Flow Diagrams (DFDs)

Data Flow Diagrams model the logical flow of data through processes, external entities, and data stores.

Level-0: Context Diagram

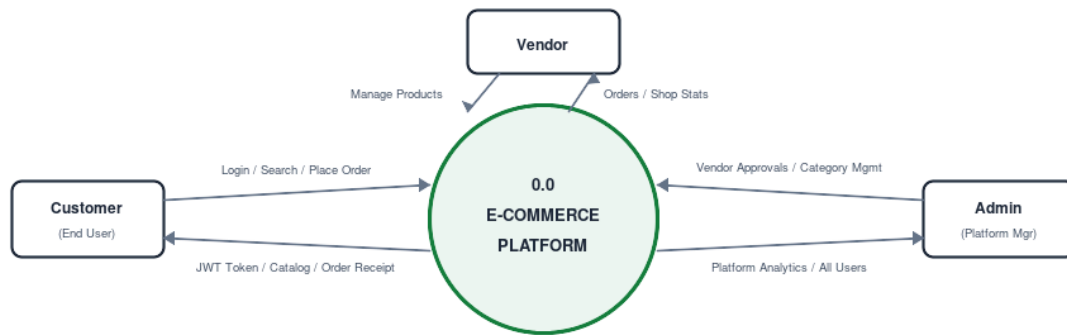


Figure 2 — Data Flow Diagram Level 0 (Context Diagram)

Level-1: Detailed Data Flow Diagram

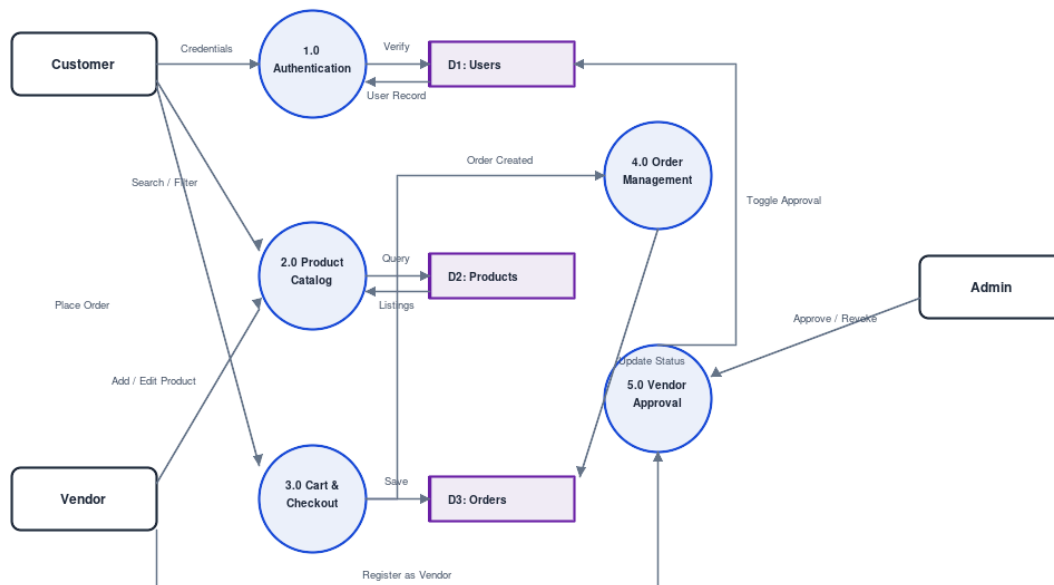


Figure 3 — Data Flow Diagram Level 1 showing internal system processes

4.5 Database Entity Overview Table

The platform's relational database consists of six core entities that store user details, vendor configurations, catalog items, and checkout history.

Entity Name	Primary Key	Foreign Keys	Purpose
User	<code>id</code> (Auto)	None	Stores credentials, roles (customer/vendor/admin), phone, and address. Email is the unique login identifier.
Vendor	<code>id</code> (Auto)	<code>user_id</code> (OneToOne → User)	Defines the vendor shop (name, description) and approval status. Auto-created on User registration with <code>role='vendor'</code> .
Category	<code>id</code> (Auto)	None	Represents product classification groups (name + unique slug). Managed by Admin only.
Product	<code>id</code> (Auto)	<code>vendor_id</code> , <code>category_id</code>	Stores items listed by vendors with name, description, price, stock, and active status.
ProductImage	<code>id</code> (Auto)	<code>product_id</code> (FK → Product)	Stores one or more uploaded images per product. Each image has an <code>is_primary</code> flag and is served via Django's <code>MEDIA_URL</code> .
Order	<code>id</code> (Auto)	<code>customer_id</code> (FK → User)	Header record with order status (pending/shipped/delivered/c

			ancelled), delivery address, total amount, and creation timestamp.
OrderItem	<code>id</code> (Auto)	<code>order_id</code> , <code>product_id</code>	Line-item snapshot of each product purchased at checkout price. Price is frozen at purchase time to preserve historical accuracy.